

CodeAlpha – Data Visualization using Power BI

Task 3: Data Visualization

This project is completed as part of the CodeAlpha Data Analytics Internship. The objective of this task is to convert raw sales data into clear and meaningful visual representations using Power BI. The dashboard created helps in understanding business performance and supports data-driven decision-making.

1. Project Objective

The main goal of this project is to analyze a sales dataset and present insights through interactive visualizations. Power BI is used to design charts and key performance indicators that highlight sales patterns, customer distribution, and regional performance.

2. Dataset Information

The dataset used for this task is train.csv. The dataset does not contain Profit or Quantity columns. The analysis and visualizations are based only on the available fields such as Order ID, Order Date, Customer ID, Category, Sub-Category, Region, Segment, and Sales.

3. Tools and Technologies

Power BI Desktop is used for creating dashboards and visualizations. DAX is used to create calculated measures for analysis.

4. Measures Created

Total Sales is calculated as the sum of the Sales column.

Total Orders is calculated by counting the number of Order ID values.

Total Customers is calculated by counting distinct Customer ID values.

Average Sales per Customer is calculated by dividing Total Sales by the number of distinct customers.

5. Visualizations Created

- a. KPI cards are used to display Total Sales, Total Orders, Total Customers, and Average Sales per Customer. These KPIs provide a quick overview of overall business performance.
- b. A bar chart is created to show Sales by Category. This visualization helps in comparing sales performance across different product categories and identifying the top-performing category.
- c. A column chart is used to display Sales by Sub-Category. This provides a more detailed comparison within each category and helps identify high and low selling sub-categories.
- d. A line chart is created to represent Sales Trend Over Time using the Order Date. This visualization helps in identifying seasonal patterns and changes in sales over time.
- e. A pie or donut chart is used to show Sales Distribution by Region. This visualization highlights the contribution of each region to total sales.
- f. A table visual is included to display Category, Sub-Category, and Sales values for detailed reference.

- g. Slicers are added for Region, Category, and Segment to make the dashboard interactive and allow users to filter the data dynamically.

6. Dashboard Overview

The dashboard is designed with KPI cards at the top for summary insights, followed by charts for comparison and trend analysis. Slicers are placed to enable easy filtering and exploration of the data.

7. Key Insights

The dashboard helps identify top-performing categories and regions, observe sales trends over time, and understand customer value through average sales per customer. These insights can support better business decisions.

8. Conclusion

This project successfully fulfills Task 3 of the CodeAlpha Data Analytics Internship. By using a different dataset and Power BI for visualization, meaningful insights are presented through an interactive and well-structured dashboard.